

T-030 conditional consolidation and external-literature positioning:

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

This is a **conditional consolidation**, not an independent proof certificate. It proves no new theorem. It (i) records the honest status of the three precise versions of T-030, (ii) pins each to its source – internal (R-025/R-026/R-032, with statements and complete proofs appended in Sec. A) or external (Mudgal 2022; Bourgain–Demeter), and (iii) on that basis terminates the analytical reduction treadmill. Every closure claim below is conditional on the cited proof, which is either the appended internal result or a textbook input named explicitly.

2. The three precise versions of T-030

$$N = |Q|, E_+(Q) = \#\{(a, b, c, d) \in Q^4 : a + b = c + d\}.$$

T-030a (lattice, R^ε -loss) – CLOSED conditionally on R-026.

$$Q \subset \mathbb{Z}^3 \cap \{|x|^2 = R\} \implies E_+(Q) \ll_\varepsilon R^\varepsilon N^2.$$

Source: R-026, assembled as the self-contained Theorem in Sec. A.4 (T7-NTstandard – a theorem modulo the textbook Dirichlet class-number formula and the divisor bound only. *External positioning:* A. Mudgal, *Additive energies on spheres*, JAMS 106(4):2927–2958 (2022), arXiv:2105.06925, studies exactly $A \subset \mathbb{Z}^d \cap \{|x|^2 = m\}$ additive energy ($d = 3, 4$); $d = 4$ gives $\ll_\varepsilon m^\varepsilon |A|^{2+1/3-1/1392}$, $d = 3$ improves Benatar–Maffucci. Mudgal confirms the object is studied and controlled, but is NOT the source of the clean $R^\varepsilon N^2$ here – that is R-026 (Sec. A.3); no specific Mudgal $d = 3$ theorem number is attributed.

T-030b (lattice, N^2 polylog) – OPEN. $Q \subset \mathbb{Z}^3 \cap \{|x|^2 = R\} \stackrel{?}{\implies} E_+(Q) \lesssim N^2 \log^{O(1)} N$: the ε -removal $R^\varepsilon \rightarrow \log^{O(1)} |Q|$, strictly stronger than T-030a, not found as a known theorem.

T-030-real (arbitrary real $Q \subset S^2$) – OPEN (Bourgain–Demeter). $Q \subset S^2 \stackrel{?}{\implies} E_+(Q) = E_2(Q) \lesssim_\varepsilon (\#Q)^{2+\varepsilon}$: a Bourgain–Demeter conjecture (exposed by Sheffer). As of Sheffer’s 2016 exposition the best reported bound for E_2 on S^2 was $O(N^{7/3})$ – cited as the 2016 state, NOT a current-best claim (the literature may have advanced). A DISTINCT object, the third-order energy E_3 on S^1 , has best-known $O(\#A^{7/2})$ (Szemerédi–Trotter).

3. The B5 closure is independent of T-030a/b/c

This is the load-bearing correction. B5’s additive-energy input is the off-diagonal floor over the ADMISSIBLE (coherence-resolved, $\theta_{\min}$ -separated) competitor class, and it is closed by TWO elementary

lemmas, using the sum-circle OCCUPANCY: By Lemma 2 (R-032, Sec. A.2) the occupancy obeys $T'(Q) \leq \lfloor 2\pi/\theta_{\min} \rfloor = 10$; by Lemma A (R-025, Sec. A.1), $E_+(Q) \leq (1 + T'(Q))N^2$. Composing,

$$E_+(Q) \leq (1 + 10)N^2 = 11N^2 = O(N^2).$$

Here $T'(Q)$ is the maximum occupancy of one sum-level circle C_m . **It is a different quantity from** (i) the few-circles COVER NUMBER L (number of circles needed to cover Q ; R-038/R-039 give $E_+ \leq 3L^2N^2$ resp. $6(L+1)N^2$), and from (ii) the lattice divisor bound (R-026). Consequently:

- B5's admissible closure needs **neither R-026 nor the cover number L** ; it is Lemma 2 + Lemma A only, giving the clean $E_+ \leq 11N^2$.
- R-026 (lattice $R^\varepsilon N^2$) is the SEPARATE statement T-030a, about the $\mathbb{Z}^3$ point class, not the admissible-competitor closure.
- R-038..R-042 (few-circles L , height multiplicity, Sidon, height energy) are structural results bearing on the OPEN real-sphere version (c); they are not part of B5's closure.

The v1.0 phrasing "the load-bearing admissible/lattice class is closed (R-026 + Lemma-2 cap)" conflated these independent arguments and is corrected here: In short, B5 = T7-SCOPE_{adm} rests on Lemma 2 + Lemma A (occupancy $T' \leq 10$); T-030a/b/c are not prerequisites.

4. Internal frontier results (this session) and what they bear on

ID	statement	tier	bears on
R-025	$E_+ \leq (1 + T')N^2$ (Lemma A)	T7	B5 closure (with R-032)
R-032	admissible $\Rightarrow T' \leq 10$ (Lemma 2)	T7	B5 closure (with R-025)
R-026	lattice $E_+ \ll_\varepsilon R^\varepsilon N^2$	T7-NTstandard	T-030a
R-038	few-circles $E_+ \leq 3L^2N^2$; witness technique-only	T7-in-class	(c) structure
R-039	linear $E_+ \leq 6(L+1)N^2$; $L \geq N^\delta/12$	T7-in-class	(c) structure
R-040	distinct-radius $E_+ \leq (2p_{\max} + 1)N^2$	T7-in-class	(c) structure
R-041	Sidon heights $E_+ \leq 6N^2$, incidence-free	T7-in-class	(c) sub-case
R-042	height-energy bound + sum-annulus mechanism	PARTIAL	(c) localisation

5. Termination of the analytical reduction treadmill (strategic)

T-030-real (E_2 on S^2) is a recognised Bourgain–Demeter OPEN PROBLEM (Sec. 2). An external dossier ran a ~ 140 -step internal reduction chain that never closed it and self-diagnosed the “analytical-closure illusion”. Methodological boundary:

- Further internal reduction chains aimed at an unconditional real-sphere $E_2 \lesssim (\#Q)^{2+\varepsilon}$ are NOT TECT deliverables (they re-derive an open research problem). Progress on (c) should be framed as a contribution to the Bourgain–Demeter problem with external incidence/decoupling machinery.
- Honest T-030 status: (a) closed conditionally on R-026; (b),(c) open in the literature; none load-bearing for B5 (Sec. 3).

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “R-043 is presented as a proof.” CORRECTED: status is CONDITIONAL CONSOLIDATION (Sec. 1); T-030a closure is conditional on R-026, whose statement and complete proof are appended (Sec. A.3 [DIV-CIRC] + Sec. A.4 the assembled T-030a theorem).

- (β) “*B5 mixes the lattice and admissible arguments.*” CORRECTED (Sec. 3): B5 = Lemma 2 + Lemma A (occupancy $T' \leq 10 \Rightarrow 11N^2$); R-026 (lattice) and the cover number L are separate and not used.
- (γ) “*The real-sphere best-known must be sourced and dated.*” ADDRESSED (Sec. 2, App. B): the E_2 -on- S^2 bound $O(N^{7/3})$ is cited as the state REPORTED BY SHEFFER IN 2016 (date-qualified, not “current best”); the DISTINCT E_3 -on- S^1 object has $O(\#A^{7/2})$ (Szemerédi–Trotter).
- (δ) “*Does terminating the treadmill cost a B5 tier?*” DISMISSED: B5 is already T7-SCOPE_{adm} via Sec. 3; the real-sphere route is a strictly stronger frontier statement, not a prerequisite.

Quantitative sanity check (mandatory). Anchors re-cross-checked (dr2_t030_consolidation_crosscheck.py): single rich circle $E_+ = 34668$ ($N = 108$); the B5 occupancy bound $E_+ \leq 11N^2$ (Lemma A at $T' = 10$); Mudgal $d = 4$ exponent $2 + \frac{1}{3} - \frac{1}{1392} \approx 2.3326$ recorded with provenance; the open E_2 -on- S^2 conjecture exponent $2(+\varepsilon)$; NO unverified best-known exponent used. No internal numeric altered.

Appendix A. Self-contained proof of T-030a (with the B5-closure lemmas)

A.1 Lemma A (R-025; T7; full proof: dr2-circle-richness-reduction v1.0). For finite $Q \subset S^2$, $E_+(Q) \leq (1 + T'(Q))N^2$ where $T'(Q) = \max_{m \neq 0, r(m) > 0} \#(Q \cap C_m)$, $C_m = S^2 \cap \{x \cdot m = |m|^2/2\}$. *Proof.* For $m \neq 0$, $a + b = m$ forces $a, b \in C_m$, so $r(m) \leq \#(Q \cap C_m) \leq T'$; hence $\sum_{m \neq 0} r(m)^2 \leq T' \sum_{m \neq 0} r(m) \leq T'N^2$, and $r(0)^2 \leq N^2$. Elementary, decoupling-free. $\square$

A.2 Lemma 2 (R-032; T7; full proof: coherence-offdiag-comfortable-bound v1.0). For a $\theta_{\min}$ -separated Q on a sphere of radius Q_0 , every proper sum-circle C_t (radius $\rho = \sqrt{Q_0^2 - |t|^2/4} \leq Q_0$) has $\#(Q \cap C_t) \leq \lfloor 2\pi/\theta_{\min} \rfloor$. *Proof.* Points on C_t are pairwise Euclidean $\geq 2Q_0 \sin(\theta_{\min}/2)$; k points pairwise $\geq d$ on a radius- ρ circle satisfy $k \leq \pi/\arcsin(d/2\rho)$, and $\rho \leq Q_0$ gives $d/2\rho \geq \sin(\theta_{\min}/2)$, so $k \leq 2\pi/\theta_{\min}$. For the TECT admissible window this is 10. $\square$

A.3 Lattice sum-circle occupancy [DIV-CIRC] (R-026; T7-NTstandard; full proof: dr2-lattice-divisor-closure v1.2). For $m \in \mathbb{Z}^3$ with $0 < |m|^2 < 4R$ (so $R' := 4R - |m|^2 > 0$),

$$\#(\mathbb{Z}^3 \cap C_m) \leq 6d(R') \ll_{\varepsilon} R^{\varepsilon},$$

uniformly in m (d the divisor function); the degenerate boundary $|m|^2 = 4R$ ($R' = 0$) gives $\#(\mathbb{Z}^3 \cap C_m) \leq 1$ (case (i) below). *Proof.* (i) *Shift removal.* C_m has centre $m/2$. For $x \in \mathbb{Z}^3 \cap C_m$ set $y := 2x - m \in \mathbb{Z}^3$; then $y \cdot m = 2x \cdot m - |m|^2 = 0$, so y lies in the rank-2 sublattice $\Lambda_m := \mathbb{Z}^3 \cap m^{\perp}$, and $|y|^2 = 4|x|^2 - 4x \cdot m + |m|^2 = 4R - |m|^2 =: R'$. The map $x \mapsto y$ is injective ($x = (y + m)/2$), so $\#(\mathbb{Z}^3 \cap C_m) \leq r_Q(R') := \#\{y \in \Lambda_m : |y|^2 = R'\}$, a HOMOGENEOUS representation count by the positive-definite integral binary quadratic form $Q(y) = |y|^2$ on Λ_m (the centre $m/2$ is removed exactly). If $R' = 0$ (i.e. $|m|^2 = 4R$, the tangent-plane degeneracy) then $y = 0$ is the only solution and, by injectivity, $\#(\mathbb{Z}^3 \cap C_m) \leq 1$; so assume henceforth $0 < |m|^2 < 4R$, whence $R' > 0$ and $d(R')$ is defined. (ii) *Class-number bound.* For a primitive positive-definite integral binary form of discriminant Δ , Dirichlet’s class-number formula gives $\sum_{[f] \text{ disc } \Delta} r_f(n) = w(\Delta) \sum_{d|n} \chi_{\Delta}(d) \leq 6d(n)$ ($w(\Delta) \leq 6$, $\chi_{\Delta} \leq 1$); a single class is bounded by the sum over all classes, so $r_Q(R') \leq 6d(R')$ (an imprimitive content g reduces to $r_{Q'}(R'/g) \leq 6d(R')$). This is UNIFORM in m (independent of Δ , hence of the plane). (iii) *Divisor bound.* $d(R') \ll_{\varepsilon} R'^{\varepsilon} \leq (4R)^{\varepsilon} \ll_{\varepsilon} R^{\varepsilon}$. $\square$ Both inputs – the class-number formula and $d(n) \ll_{\varepsilon} n^{\varepsilon}$ – are textbook, so [DIV-CIRC] is a theorem, not a hypothesis.

A.4 Theorem (T-030a), self-contained modulo textbook number theory. For $Q \subset \mathbb{Z}^3 \cap \{|x|^2 = R\}$,

$$E_+(Q) \leq (1 + C_{\varepsilon} R^{\varepsilon})N^2 \ll_{\varepsilon} R^{\varepsilon} N^2.$$

Proof. By A.1 (Lemma A), $E_+(Q) \leq (1 + T'(Q))N^2$ with $T'(Q) = \max_{m \neq 0, r(m) > 0} \#(Q \cap C_m)$. Each contributing $m = a + b$ ($a, b \in Q$) has $0 < |m|^2 \leq 4R$; by A.3 its occupancy is ≤ 1 when $|m|^2 = 4R$ and $\leq 6d(4R - |m|^2)$ when $0 < |m|^2 < 4R$, so

$$T'(Q) \leq \max \left\{ 1, 6 \max_{0 < |m|^2 < 4R} d(4R - |m|^2) \right\} \ll_\varepsilon R^\varepsilon.$$

Substituting gives the claim. $\square$ This is the frozen in-repo certificate for T-030a: it uses Lemma A (A.1) and the two textbook number-theoretic inputs of A.3 – no decoupling, no incidence conjecture, no external theorem. (Mudgal 2022 corroborates the object but is not used in this proof.)

Appendix B. References

[1] A. Mudgal, *Additive energies on spheres*, J. London Math. Soc. 106(4):2927–2958 (2022); arXiv:2105.06925. (Lattice $\mathbb{Z}^3/\mathbb{Z}^4$ sphere additive energy; $d = 4$ threshold-breaking; corroborates T-030a's object.)

[2] A. Sheffer, *Additive Energy of Real Point Sets*, expository note (adamsheffer.wordpress.com, 2016): states the Bourgain–Demeter problem $E_2(A) \lesssim_\varepsilon (\#A)^{2+\varepsilon}$ for $A \subset S^2$ (OPEN), reports the then-best $O(N^{7/3})$ for E_2 on S^2 (2016 state), and records the DISTINCT E_3 -on- S^1 best-known $O(\#A^{7/2})$ via Szemerédi–Trotter.

[3] J. Bourgain, C. Demeter, decoupling/additive-energy works (the conjecture that T-030-real instantiates; Mudgal [1] "makes progress towards one of their conjectures").

Result footer

Result ID:	R-043 (T-030 CONDITIONAL consolidation & literature positioning) / T-030 frontier. NOT a proof certificate.
Precise statement:	(a) lattice R^ε -loss CLOSED conditionally on R-026 (T7-NTstandard, complete proof appended Sec.A.3–A.4; Mudgal 2022 corroborates the object). (b) lattice N^2 polylog OPEN. (c) arbitrary real Q in S^2 , $E_2 \ll_\varepsilon (\#Q)^{2+\varepsilon}$, OPEN (Bourgain–Demeter; Sheffer 2016 reports E_2/S^2 best $O(N^{7/3})$; the distinct E_3/S^1 best is $O(\#A^{7/2})$). B5 closure is INDEPENDENT: Lemma 2 (R-032, $T' \leq 10$) + Lemma A (R-025) $\Rightarrow E_+ \leq 11N^2$; needs neither R-026 nor the few-circles cover number L.
Scope:	T-030 (H-NONLATTICE-REMAINDER). B5 = T7-SCOPE_{adm} UNAFFECTED (Sec.3). (b),(c) NON-LOAD-BEARING.
Evidence grade:	CONDITIONAL CONSOLIDATION (no new theorem). Internal lemmas R-025/R-032 T7, R-026 T7-NTstandard (statements + complete proofs appended, Sec.A). (b),(c) OPEN.
Reproduction command:	python3 codes/vacuum/dr2_t030_consolidation_crosscheck.py
Falsifier:	a counterexample to R-026 (reopens (a)); an unconditional published proof of (c) (resolves Bourgain–Demeter).
Tier before / after:	NONE -- consolidation only. No gate/tier/hypothesis flip. B5 stays T7-SCOPE_{adm}; T-030(b),(c) OPEN.
No-overclaim statement:	NOT an independent proof; T-030a is conditional on R-026 (appended). B5 closure uses Lemma 2 + Lemma A (occupancy $T' \leq 10$), not R-026 and not the cover number L. No E_2/S^2 best-known exponent is claimed. External 140-step chain NOT adopted.
Next required action:	Analytical T-030 frontier settled. Redirect to the numerical production run (strategy/numerical-programme-status-and-plan): P2 continuum-limit background (≥ 4 resolutions) $\rightarrow$ P3 Hessian/zero-modes $\rightarrow$ P4 CKM/PMNS.